

Digital Transmission System

Final Project Report - TLC Virtual Lab #2

PAM-M modulation, AWGN channel modeling, matched-filter validation,
and single-pole receiver optimization

Modulations	PAM-2, PAM-4, PAM-8, PAM-16
Pulse shapes	NRZ and Square Root Raised Cosine
Channel	Additive White Gaussian Noise
Receiver cases	Matched filter and single-pole low-pass filter
Target metric	BER = 10^{-3} versus E_b/N_0

Executive Summary

This report presents a numerical simulator for a baseband digital transmission system using PAM-M modulation over an AWGN channel. The simulator validates matched-filter performance against theoretical BER curves, evaluates NRZ and SRRC pulse shaping in the spectral and time domains, and studies how a practical single-pole receiver filter changes the E_b/N_0 required to reach $BER = 10^{-3}$.

- Matched-filter simulations follow the expected theoretical PAM-M BER trend.
- NRZ and SRRC pulse shaping are compared through PSD and eye-diagram evidence.
- The single-pole receiver introduces ISI and noise-bandwidth tradeoffs, creating an optimum bandwidth for each tested case.
- For SRRC/PAM-16, the single-pole receiver remains ISI-limited in the tested sweep and does not reach $BER = 10^{-3}$.

1. System Model

The simulated chain follows the TLC Lab 2 structure: random bit generation, Gray-coded PAM-M symbol mapping, pulse shaping, AWGN addition, receiver filtering, optimum sampling, nearest-level decision, demapping, and error counting. Frequencies are normalized to the symbol rate R_s and time is normalized to the symbol interval.

For each modulation order M , $\log_2(M)$ bits form one symbol. The PAM alphabet is antipodal and normalized to unit average symbol energy. The noise variance is computed from E_b/N_0 using $E_b = E_s/\log_2(M)$ and $N_0 = E_b/(E_b/N_0)$, with real AWGN standard deviation $\sqrt{N_0/2}$.

2. Digital Filter Synthesis

The lab requires evidence that the digital filter synthesis is accurate in the signal band. The SRRC pulse is synthesized by frequency sampling the square root of the raised-cosine spectrum and applying a finite-length time-domain window. The single-pole receiver is synthesized by impulse invariance, producing $H(z) = (1-a)/(1-az^{-1})$, where $a = \exp(-2\pi f_{3dB}/S_p S)$.

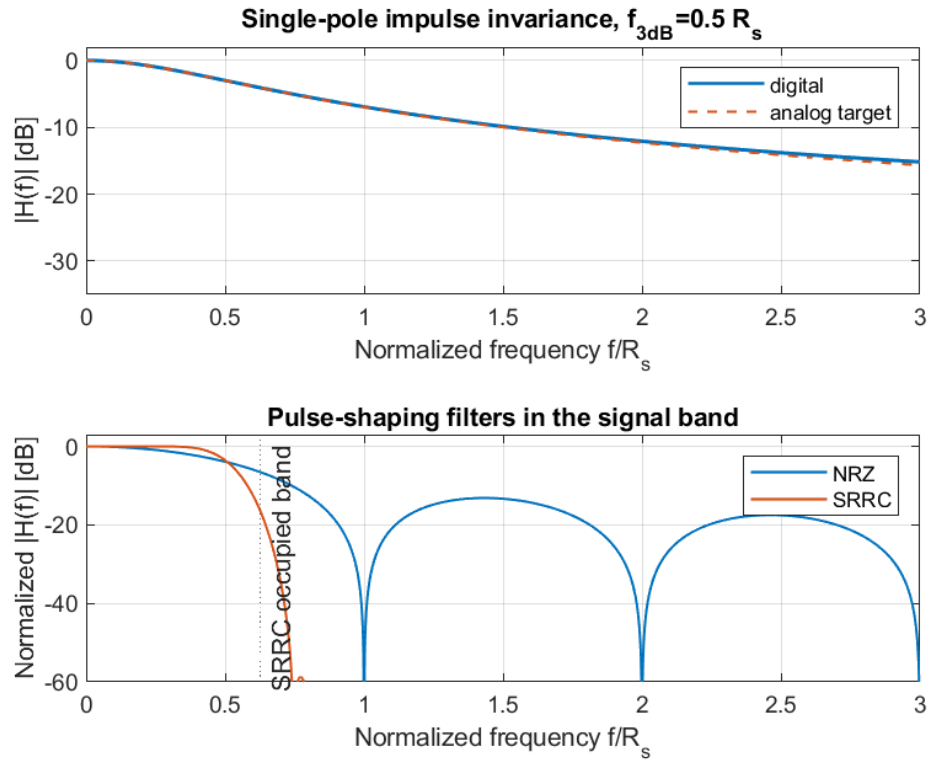


Figure 1. Digital synthesis evidence for the single-pole filter and pulse-shaping filters.

3. Spectral Analysis

The PSD comparison verifies that the selected pulse shape controls the occupied bandwidth. NRZ is compact in implementation but has sinc-like spectral sidelobes, while SRRC is designed to confine the main occupied region according to the roll-off factor.

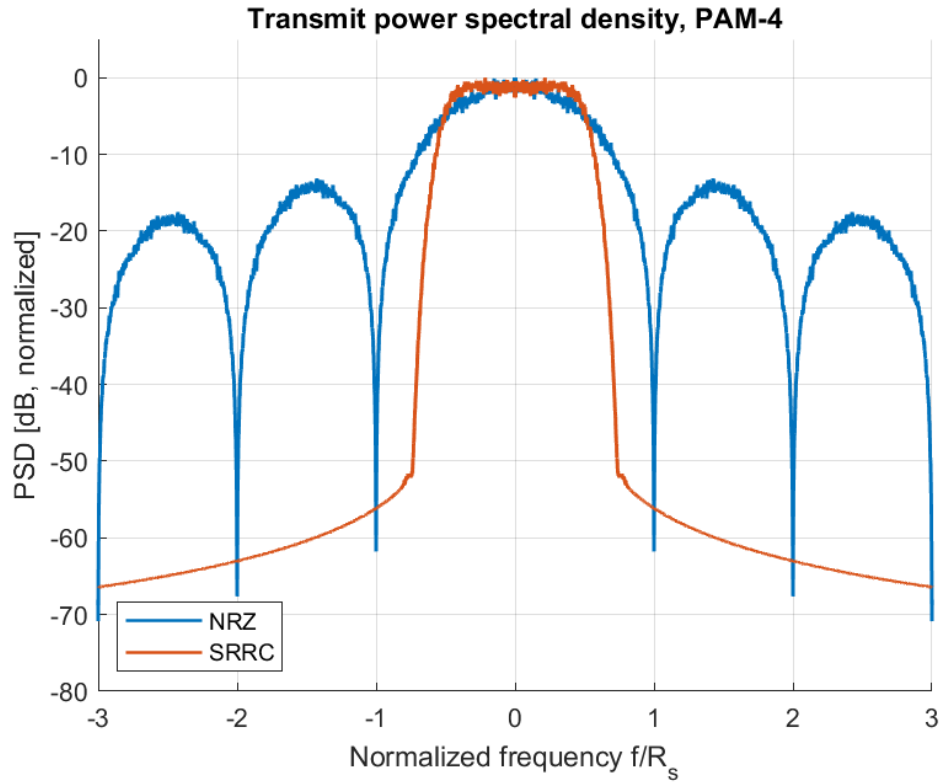


Figure 2. Normalized transmit PSD for PAM-4 using NRZ and SRRC pulses.

4. Matched-Filter Validation Against Theory

The matched-filter receiver uses the time-reversed transmit pulse. This is the reference case because it maximizes sampled SNR in AWGN when the pulse model is known. BER curves are compared against the standard Gray-coded PAM-M theoretical approximation.

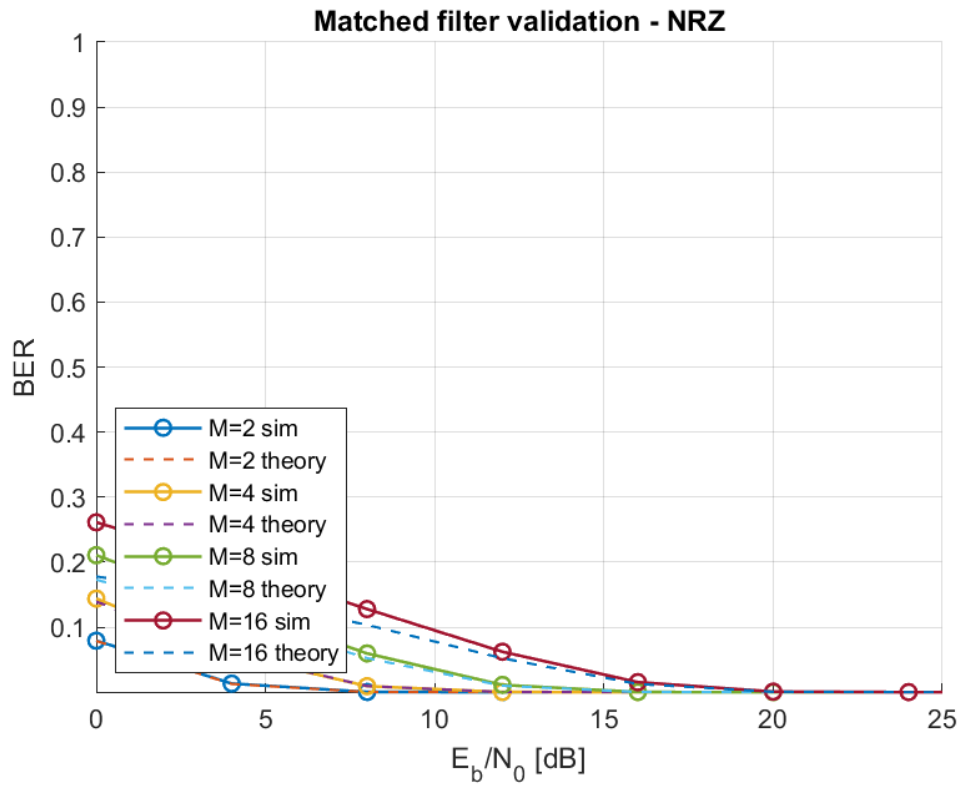


Figure 3. NRZ matched-filter BER validation for PAM-2, PAM-4, PAM-8, and PAM-16.

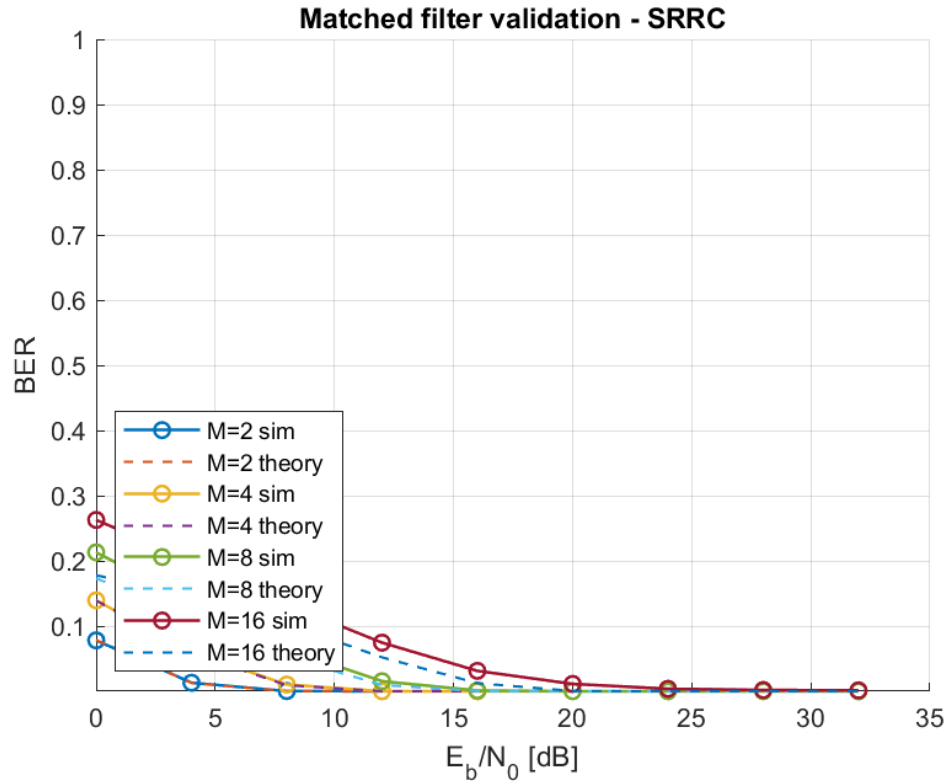


Figure 4. SRRC matched-filter BER validation for PAM-2, PAM-4, PAM-8, and PAM-16.

The simulated points track the theoretical curves with the expected Monte Carlo scatter, especially at low BER where fewer errors are available for counting. Increasing the simulated bit count further would reduce variance in the lowest-BER region.

5. Eye-Diagram Evidence

Eye diagrams are included for the noiseless waveform and for the E_b/N_0 value corresponding to $BER = 10^{-3}$. PAM-4 is used as the representative visualization case because it shows multiple decision levels while remaining readable on the page.

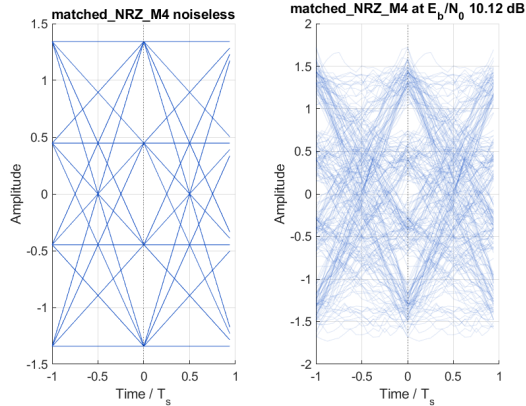


Figure 5. Matched-filter NRZ/PAM-4 eye diagrams.

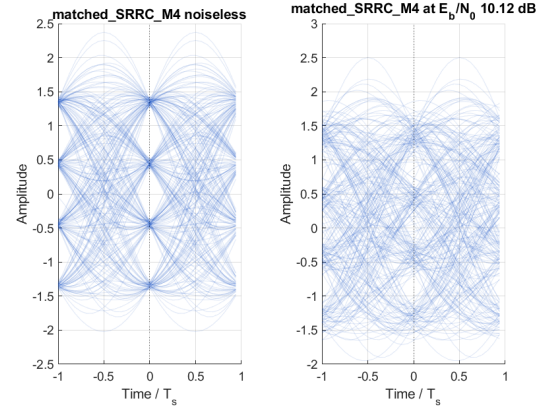


Figure 6. Matched-filter SRRC/PAM-4 eye diagrams.

6. Single-Pole Receiver Bandwidth Optimization

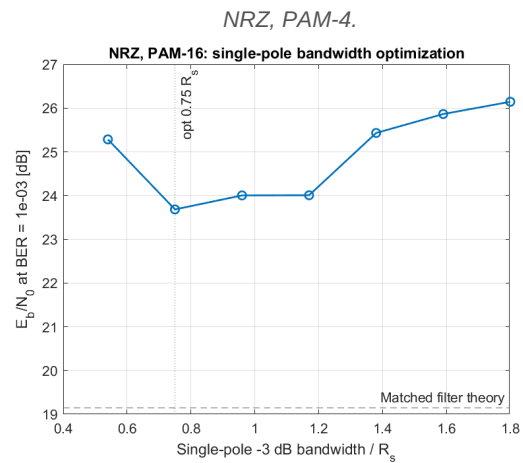
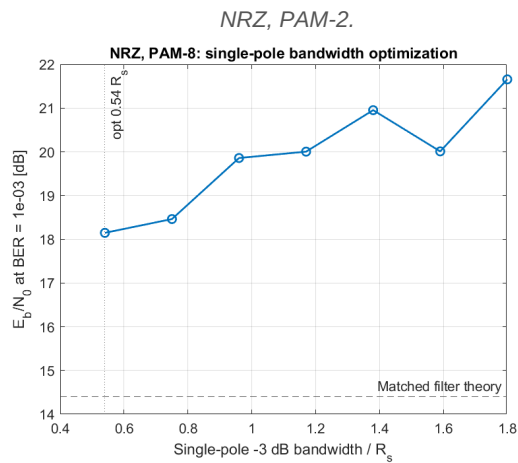
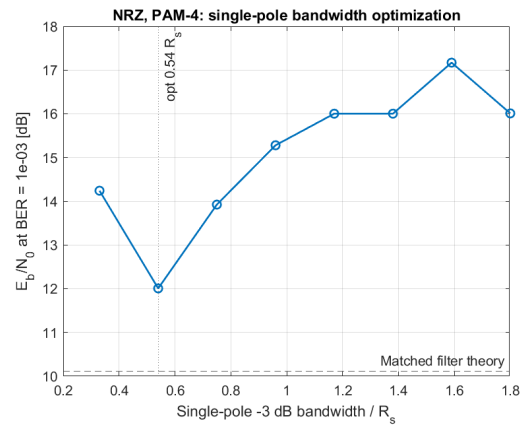
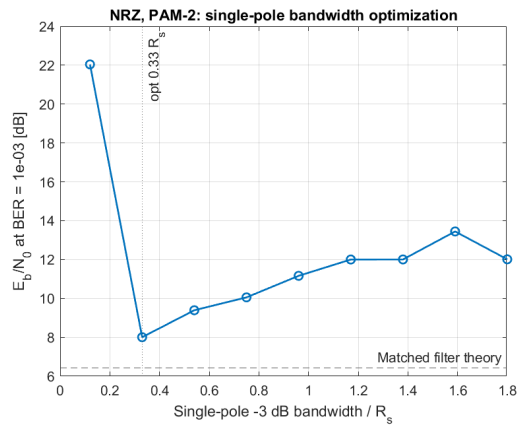
The second experiment replaces the matched filter with a practical single-pole low-pass receiver. The -3 dB bandwidth is swept, and for each bandwidth the BER curve is interpolated to estimate the E_b/N_0 required for $BER = 10^{-3}$. This exposes the tradeoff between noise rejection at low bandwidth and inter-symbol interference at poorly matched bandwidths.

Pulse	M	Best BW/Rs	Best E_b/N_0 dB	Matched dB	Penalty dB
NRZ	2	0.33	8.01	6.42	1.59
NRZ	4	0.54	12.01	10.12	1.89
NRZ	8	0.54	18.15	14.39	3.75
NRZ	16	0.75	23.68	19.15	4.54
SRRC	2	0.54	8.01	6.42	1.59
SRRC	4	0.54	13.26	10.12	3.14
SRRC	8	0.54	18.60	14.39	4.21
SRRC	16	Not reached	Not reached	19.15	Not reached

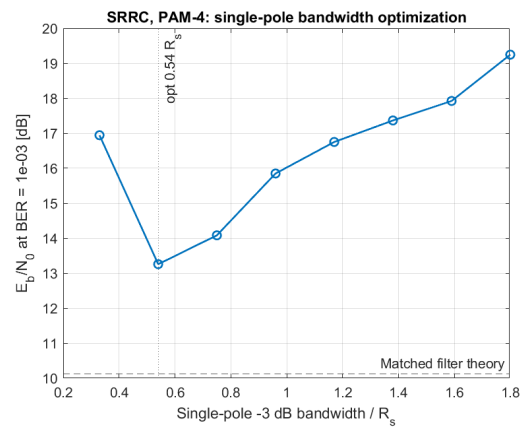
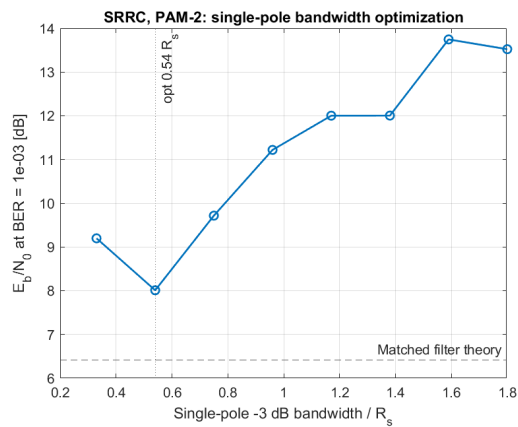
The penalty column is measured relative to the matched-filter theoretical E_b/N_0 at the same target BER. The SRRC/PAM-16 case is reported as not reached because the focused extended sweep showed an error floor above 10^{-3} for the tested single-pole receiver.

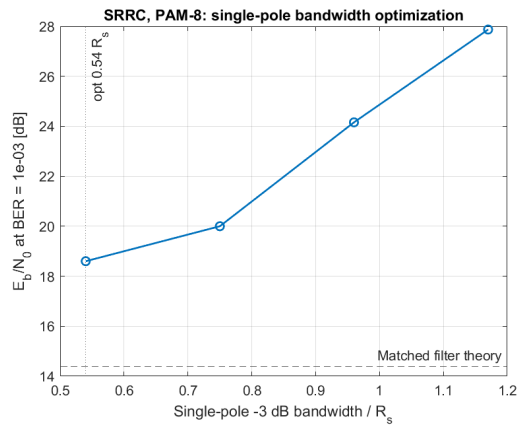
Bandwidth Sweep Plots

NRZ pulse shaping

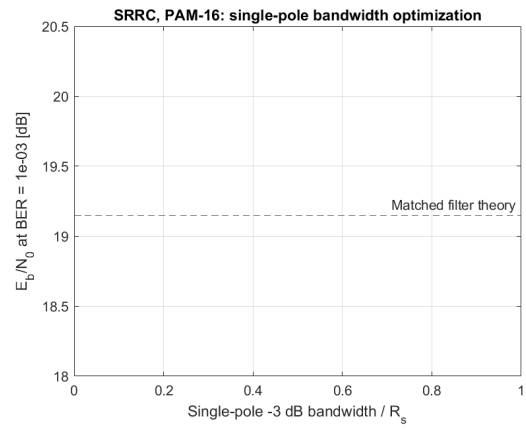


SRRC pulse shaping



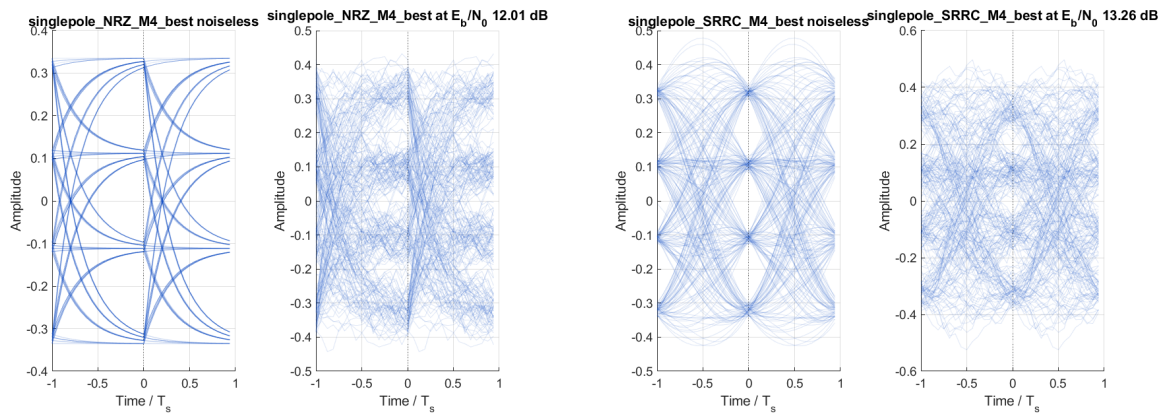


SRRC, PAM-8.



SRRC, PAM-16.

Optimized Single-Pole Eye Diagrams



Optimized single-pole NRZ/PAM-4 eye diagrams.

Optimized single-pole SRRC/PAM-4 eye diagrams.

7. Conclusions

- The simulator satisfies the TLC Lab 2 matched-filter validation requirement for NRZ and SRRC shaping.
- The spectral plots provide direct evidence of digital synthesis quality in the occupied signal band.
- The single-pole receiver has a measurable E_b/N_0 penalty relative to the matched filter because it is not matched to the transmit pulse.
- The optimum single-pole bandwidth depends on both pulse shape and PAM order, confirming that receiver bandwidth cannot be selected independently of modulation and shaping.
- The modular project structure separates configuration, filtering, modulation, simulation, plotting, and utility logic so the experiment can be extended cleanly.

Appendix: Reproducibility

Main project execution: run `TLC_Lab2_Project.m` from MATLAB. Report figure generation: run `documentation/generate_report_figures.m`. The report uses figures saved under `documentation/figures` and the optimization summary saved as `documentation/optimization_summary.csv`.